

Weather Dashboard - Interview Challenge for Atobic

Overview

Create a simple weather dashboard prototype that allows users to search for weather information by city name. This project aims to evaluate rudimentary full-stack development skills.

Requirements

Frontend

- Create a responsive single-page application with:
 - A search input for city names
 - Display current weather consisting of :
 - Temperature
 - Weather conditions (sunny, cloudy, etc.)
 - Humidity
 - Wind speed
 - A list of previously searched cities by using browser side storage
 - Basic error handling for invalid cities

Backend

- Create a simple backend API that:
 - Handles requests from the frontend
 - Integrates with a free weather API (like OpenWeatherMap's free tier)
 - Implements error handling

Technical Requirements

- Frontend Deployment: GitHub Pages or any option of your choice.
- Backend Deployment: Replit.com (Free Tier) or any option of your choice.
- Store front-end and back-end code in a public GitHub repository

Backend Deployment Instructions

For Replit :

1. Create a free Replit account
2. Create a new Node.js repl
3. Import your project or start from scratch
4. Use the "Run" button to deploy

Note: Replit will put your server to sleep after inactivity, which is acceptable for demonstration purposes. Include wake-up time expectations in your documentation.

Bonus Points

- Unit tests using Jest
- GitHub Actions for CI/CD
- Loading states and animations
- Responsive design for mobile devices
- Handle API sleep/wake gracefully with user feedback

Evaluation Criteria

1. Code organization and structure
2. Error handling and edge cases
3. UI/UX design decisions
4. API security considerations
5. Testing approach

Submission Guidelines

1. Create a public GitHub repository
2. Include a detailed README.md with:
 - Project overview
 - Setup instructions
 - API documentation
 - Design decisions
 - Future improvements
 - Notes about platform limitations (e.g., cold starts)
3. Deploy the application
4. Share the 1) Github repository URL b) Deployed application link c) Resume to talent@atobic.com